CIE AS PHYSICS (9702) • MARK SCHEME & MODEL ANSWERS

6.1 Stress & Strain

Syllabus Section: 6. Deformation of Solids **Paper Type:** Structured Questions (Paper 2) **Total:** 8 Questions (99 Marks)

Question 1 Easy [11 marks]

Part (a) [3 marks]

Forces can change the motion of a body, or cause deformation.

State

(i) the definition of *deformation*.

[1]

(ii) the name of the force which stretches an object.

[1]

(iii) the name of the force which squashes an object.

[1]

Mark Scheme (3 marks):

(i) Deformation is defined as:

- Changing the size / shape (of an object); *[1 mark]*

(ii) The force which stretches an object is called:

- Tensile (stress / force); *[1 mark]*

(iii) The force which squashes an object is called:

- Compressive (stress / force); *[1 mark]*

[Total: 3 marks]

Part (b) [2 marks]

Hooke's Law can be applied to objects which are stretched or compressed.

State the equation for Hooke's Law, defining any terms used.

Mark Scheme (2 marks):

Hooke's Law can be written as:

- $F = kx$ *[1 mark]*
- Where F = force, k = spring constant and x is the extension; *[1 mark]*

[Total: 2 marks]

Part (c) [3 marks]

A spring is suspended from a stand as shown in Fig. 1.1 and has a load of 5.0 N applied to it. The spring extends by 4.2 cm.

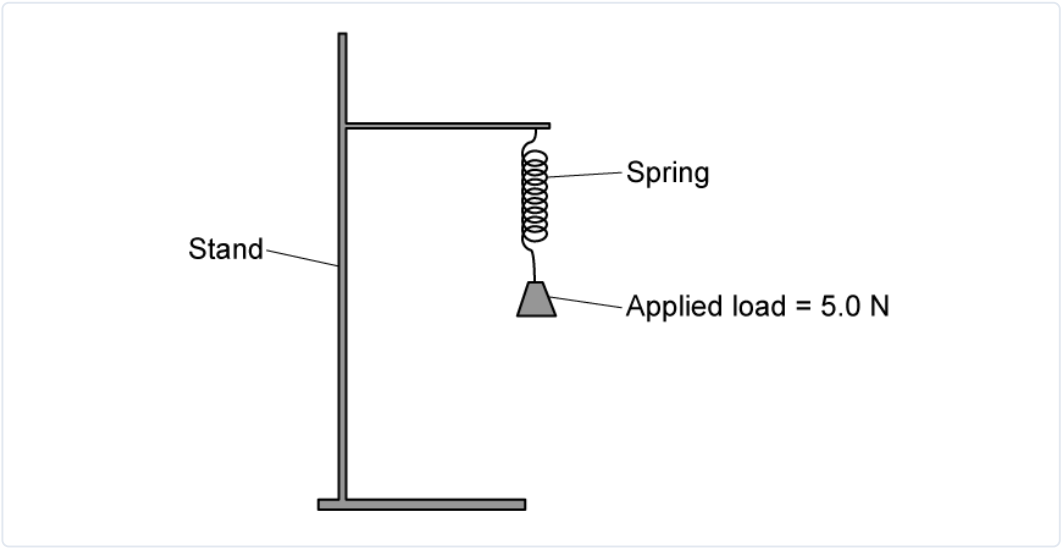


Fig. 1.1

Calculate the spring constant, expressing your answer in S.I. units.

Mark Scheme (3 marks):

Calculate the spring constant, expressing your answer in S.I. units:

List the known quantities:

- Load, F = 5.0 N
- Extension, x = 4.2 cm = 0.042 m *[1 mark]*

Use the equation for the spring constant and calculate:

- Hooke's law: $F = kx \Rightarrow k = \frac{F}{x}$
- Spring constant: $k = \frac{5.0}{0.042}$ *[1 mark]*
- Spring constant: $k = 120 \text{ N m}^{-1}$ 2 s.f. *[1 mark]*

[Total: 3 marks]

Part (d) [3 marks]

Another, thicker spring is attached to a base so that it stands upright, as shown in Fig. 1.2.

A force of 6.3 N pushes down on the spring which compresses it by 2.4 cm.

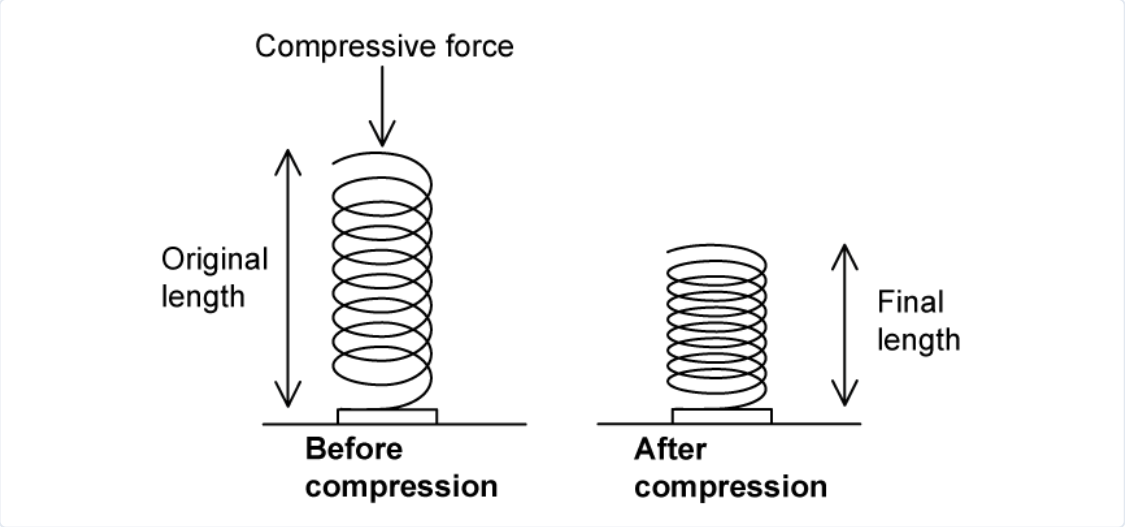


Fig. 1.2.

Calculate the spring constant, expressing your answer in S.I. units.

Mark Scheme (3 marks):

Calculate the spring constant, expressing your answer in S.I. units:

List the known quantities:

- Compressive force, F = 6.3 N
- Compression, x = 2.4 cm = 0.024 m *[1 mark]*

Use the equation for the spring constant and calculate:

- Spring constant: $k = \frac{F}{x} = \frac{6.3}{0.024}$ *[1 mark]*
- Spring constant: $k = 260 \ \text{Nm}^{-1}$ 2 s.f. *[1 mark]*

[Total: 3 marks]

Hooke's Law questions are nearly always asked about springs under tension, being stretched. However, be aware that exactly the same equation also applies to compressive forces and that it's used in the same way.

Question 2 Easy [15 marks]

Part (a) [3 marks]

Define

(i) tensile stress

[1]

(ii) tensile strain

[1]

(iii) the Young modulus.

[1]

Mark Scheme (3 marks):

(i) Tensile stress is defined as:

- The ratio of the force to the cross-sectional area

  OR

- Stress = $\frac{F}{A}$, where F = force and A = cross-sectional area; *[1 mark]*

(ii) Tensile strain is defined as:

- The ratio of extension to initial length;

  OR

- Strain = $\frac{x}{L}$, where x = extension and L = original length; *[1 mark]*

(iii) The Young modulus is defined as:

- The ratio of stress to strain;

  OR

- Young modulus = $\frac{stress}{strain} = \frac{FL}{Ax}$ *[1 mark]*

[Total: 3 marks]

Part (b) [4 marks]

Fig. 1.1 shows stress-strain graphs for two different metal wires X and Y.

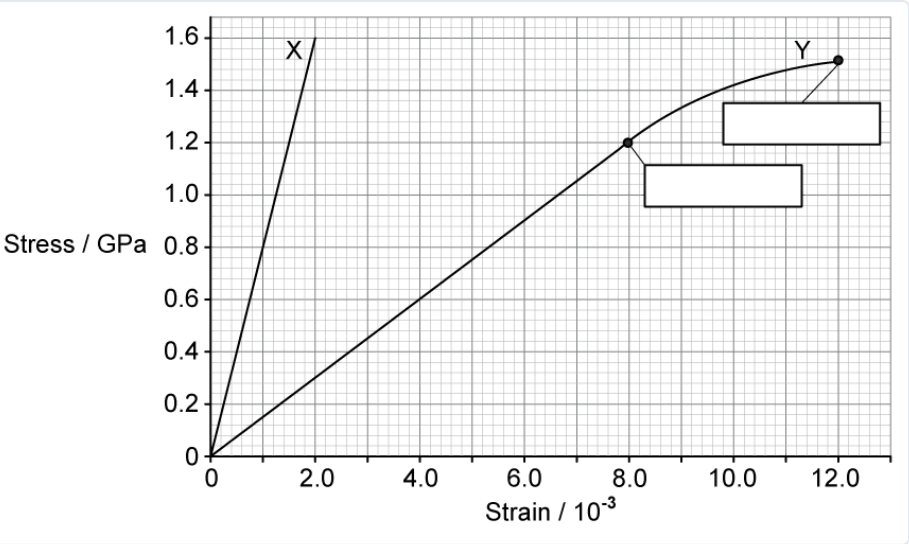
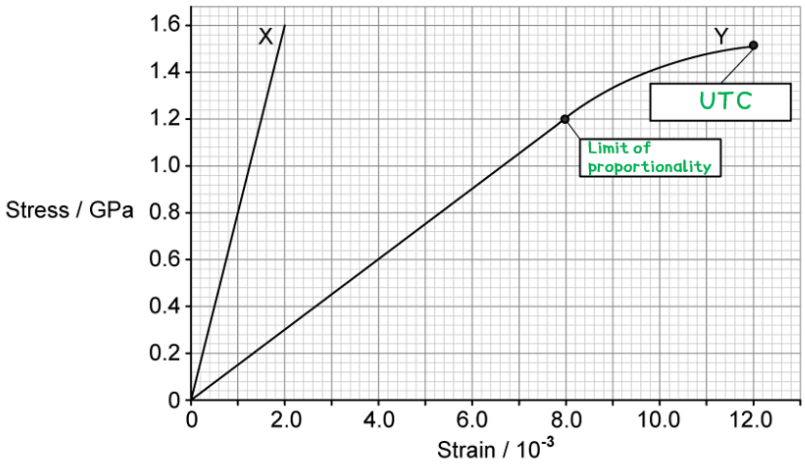


Fig. 1.1

Fill in the missing labels on Fig. 1.1 and state the definitions of these terms.

Mark Scheme (4 marks):

The correct labels are:



- Limit of proportionality (as shown); *[1 mark]*
- Ultimate tensile strength / UTS / breaking stress (as shown); *[1 mark]*

Limit of proportionality is defined as:

- The point beyond which the wire no longer obeys Hooke's law; *[1 mark]*

Ultimate tensile strength / breaking stress is defined as:

- The maximum stress the wire is able to support / before it breaks; *[1 mark]*

[Total: 4 marks]

Part (c) [2 marks]

State and explain which wire, X or Y, has the greater Young modulus.

Mark Scheme (2 marks):

The wire with the greater Young modulus is

- Wire X; *[1 mark]*

Because:

- X has the steeper gradient / larger value of stress/strain (before breaking); *[1 mark]*

[Total: 2 marks]

Part (d) [6 marks]

Use the graph in Fig. 1.1 to determine

(i) the Young modulus of wire X and state the unit.

[3]

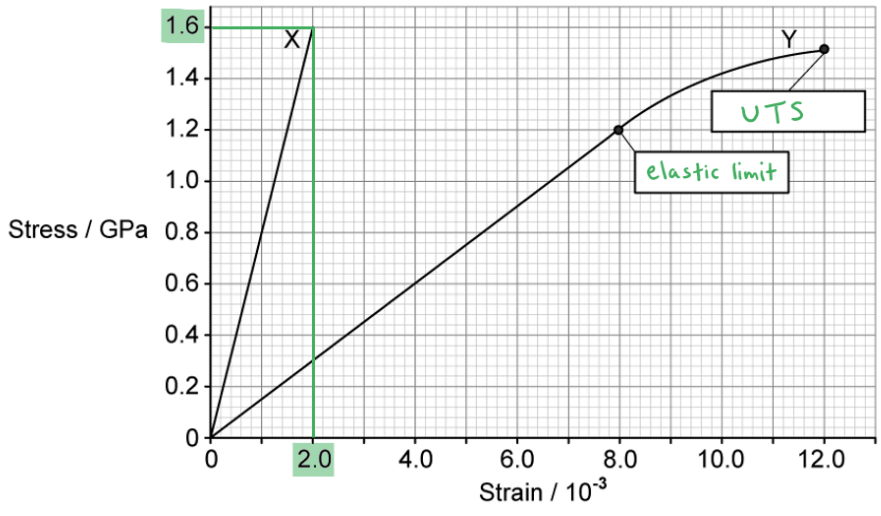
(ii) the extension of wire Y when the stress is 1.2 GPa, and state the unit.

original length of wire Y = 2.0 m

[3]

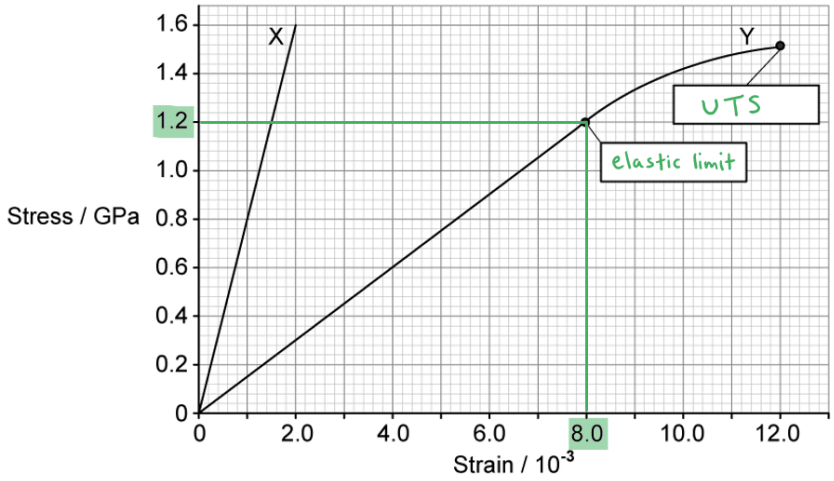
Mark Scheme (6 marks):

(i) Determine the Young modulus of wire X:



- (From graph) when stress = 1.6 GPa, strain = 2.0×10^{-3} *[1 mark]*
- Young modulus = gradient of the stress-strain graph
- Young modulus = $\frac{stress}{strain} = \frac{1.6 \times 10^{9}}{2.0 \times 10^{-3}}$ *[1 mark]*
- Young modulus = 8.0×10^{11} Pa **OR** 800 GPa *[1 mark]*

(ii) Determine the extension of wire Y when the stress is 1.2 GPa:



- (From graph) when stress = 1.2 GPa, strain = 8.0×10^{-3} *[1 mark]*
- Original length, L = 2.0 m
- Extension: $\varepsilon = \frac{x}{L} \Rightarrow x = \varepsilon L$
- Extension: $x = 8.0 \times 10^{-3} \times 2.0$ *[1 mark]*
- Extension: x = 0.016 m **OR** 1.6 cm **OR** 16 mm *[1 mark]*

[Total: 6 marks]

Question 3 Easy [11 marks]

Part (a) [5 marks]

A student is carrying out an experiment to determine the Young modulus of a wire. The student sets up the apparatus as shown in Fig. 1.1.

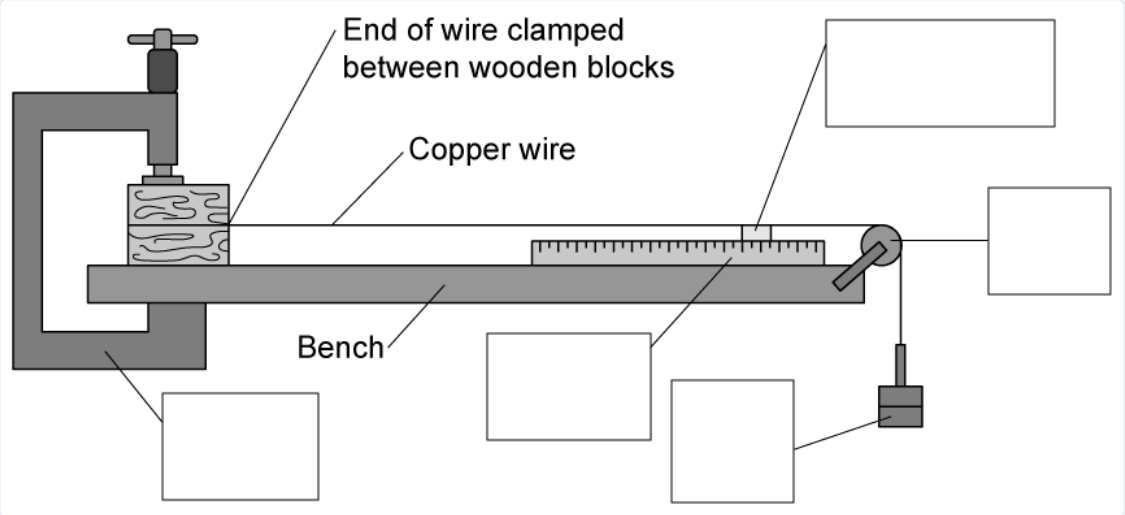
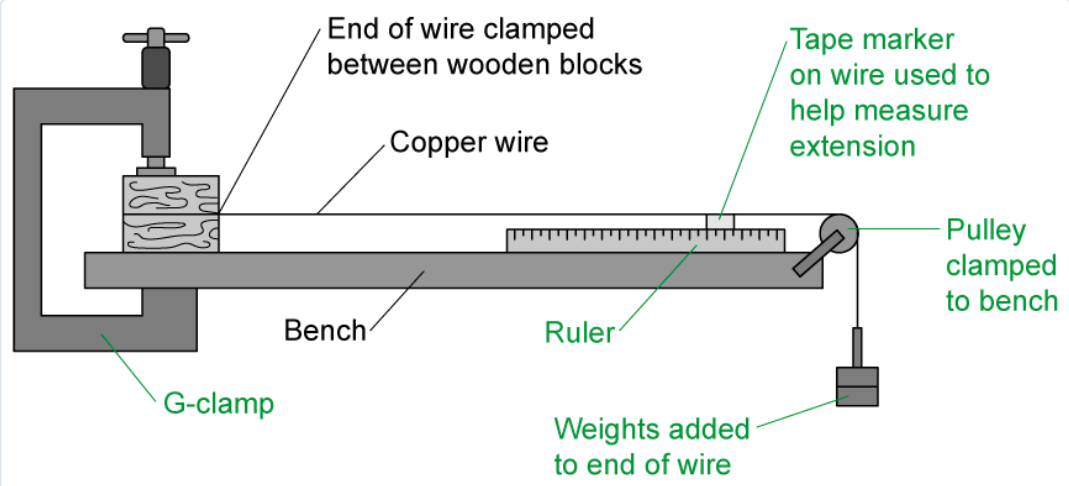


Fig. 1.1

Fill in the missing labels for the apparatus in Fig. 1.1.

Mark Scheme (5 marks):

The correct labels are as follows:



- Tape marker / marker; *[1 mark]*
- Pulley; *[1 mark]*
- G–clamp; *[1 mark]*
- Ruler; *[1 mark]*

- Weights / masses; *[1 mark]*

  One mark for each correct label

[Total: 5 marks]

Part (b) [2 marks]

State and explain a safety consideration for this experiment.

Mark Scheme (2 marks):

A safety consideration for this experiment is:

EITHER

- Wearing safety goggles at all times; *[1 mark]*
- Explanation: in case the wire snaps; *[1 mark]*

OR

- Placing a cushion / mat directly below the mass hanger; *[1 mark]*
- Explanation: in case the masses fall onto the ground / student's foot; *[1 mark]*

[Total: 2 marks]

Part (c) [2 marks]

On the axis of Fig. 1.2 below, draw a graph to show the expected variation of strain with stress for this experiment.

Assume that the wire does not stretch beyond the limit of proportionality.

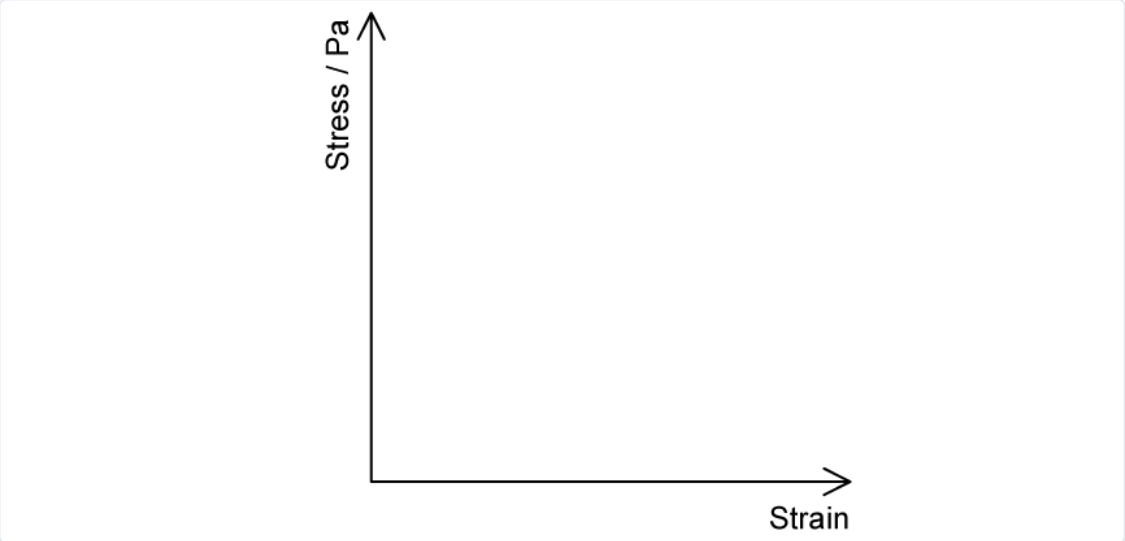
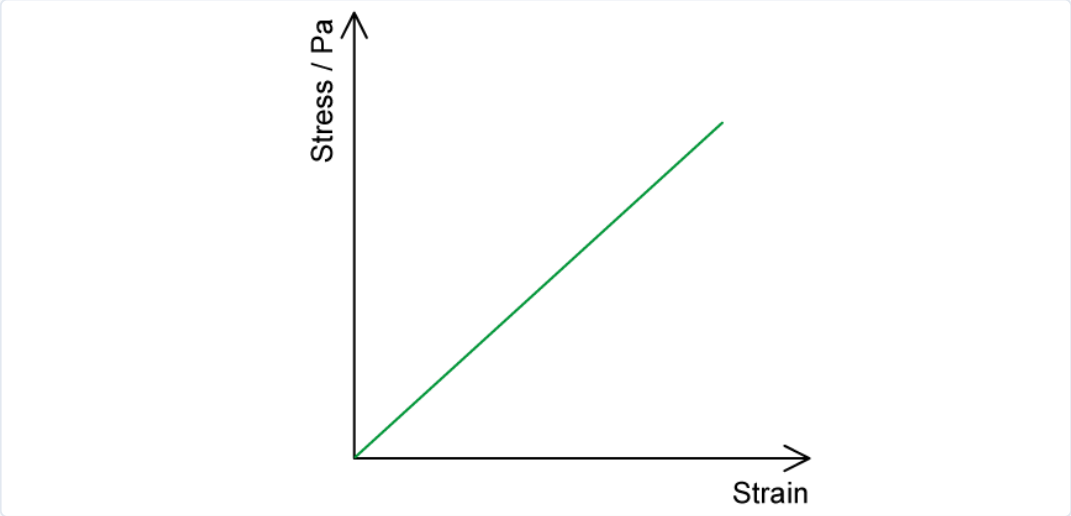


Fig. 1.2

Mark Scheme (2 marks):

The expected variation of strain with stress for this experiment is as follows:



- Straight line drawn; *[1 mark]*
- Through the origin; *[1 mark]*

[Total: 2 marks]

*The question states that the wire is not stretched beyond the limit of proportionality. This means that **Hooke's law** is **obeyed**, therefore the stress will be **directly proportiona**l to the strain. This means the graph will be a **straight line** through the **origin**.*

Remember to use a ruler or straight edge for any lines, messy graphs will often not get full marks!

Part (d) [2 marks]

Explain how the student could use the graph in Fig. 1.2 to determine the Young modulus of the wire.

Mark Scheme (2 marks):

The student could use the graph in Fig. 1.2 to determine the Young modulus because:

- Young modulus = $\frac{stress}{strain}$ *[1 mark]*
- Therefore, the gradient of the stress-strain graph = Young modulus; *[1 mark]*

[Total: 2 marks]

Question 4 Medium [12 marks]

Part (a) [2 marks]

(i) Define force.

[1]

(ii) State what is meant by work done.

[1]

Mark Scheme (2 marks):

(i) Force is defined as:

- The rate of change of momentum **OR** change in momentum ÷ time taken; *[1 mark]*

(ii) Work done is defined as:

- Force × Displacement in the direction of the force; *[1 mark]*

[Total: 2 marks]

Part (b) [10 marks]

A block of mass 0.40 kg slides in a straight line with a constant speed of 0.30 m s^{-1} along a horizontal surface, as shown in Fig. 3.1.

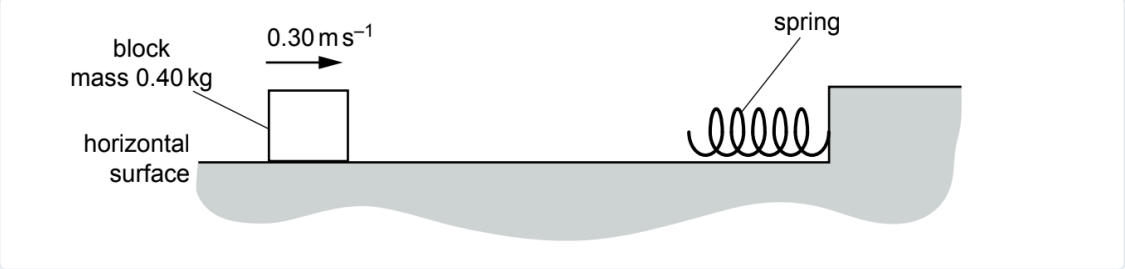


Fig. 3.1

Assume that there are no resistive forces opposing the motion of the block.

The block hits a spring and decelerates. The speed of the block becomes zero when the compression of the spring is 8.0 cm.

(i) Calculate the initial kinetic energy of the block.

kinetic energy = ... J [2]

(ii) The variation of the compression x of the spring with the force F applied to the spring is shown in Fig. 3.2.

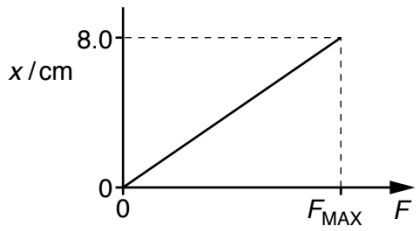


Fig. 3.2

Assume that the elastic potential energy of the spring when its compression is 8.0 cm is equal to the initial kinetic energy of the block.

Use your answer in **(b)(i)** to calculate the maximum force F_{max} exerted on the spring by the block.

F_{max} = ... N [2]

(iii) Calculate the maximum deceleration of the block.

deceleration = ... m s^{-2} [2]

(iv) State and explain whether the block is in equilibrium:

- before it hits the spring

.....

- when its speed becomes zero.

.....

[2]

(v) The block is now replaced by another block of the same mass. Frictional forces affect the motion of this block so that it has a speed of 0.25 m s^{-1} when it makes contact with the spring.

A short time later, the block has a speed of 0.15 m s^{-1} as it loses contact with the spring and moves back along its original path.

Calculate the magnitude of the change in momentum of the block.

change in momentum = ... N s [2]

Mark Scheme (10 marks):

List the known quantities:

- Mass of block, m = 0.40 kg
- Velocity of block, v = 0.30 m s^{-1}

(i) Calculate the initial kinetic energy of the block:

- $E_k = \frac{1}{2}mv^2$ *[1 mark]*

- $E_k = \frac{1}{2} \times 0.4 \times 0.3^2$
- $E_k = 0.018$ J *[1 mark]*

(ii) Calculate the maximum force F_{max} exerted on the spring by the block:

From the graph when $F = F_{max}$:

- Compression, x = 8.0 cm = 8 × 10^{-2} m

Equate the elastic potential E_p and E_k of the block and rearrange for F_{max}:

- $E_k = E_p = \frac{1}{2} F_{max} x$
- $0.018 = \frac{1}{2} \times F_{max} \times 8 \times 10^{-2}$ *[1 mark]*
- $F_{max} = \frac{0.018 \times 2}{8 \times 10^{-2}}$
- $F_{max} = 0.45$ N *[1 mark]*

(iii) Calculate the maximum deceleration of the block:

- $F = ma$ *[1 mark]*
- $a = \frac{F_{max}}{m}$
- $a = \frac{0.45}{0.40}$
- $a = 1.125$ m s^{-2} = 1.1 m s^{-2} 2 s.f. *[1 mark]*

(iv) Consider whether the block is in equilibrium during the following:

Before it hits the spring:

- The velocity is constant / the resultant force is zero, so the block is in equilibrium; *[1 mark]*

When its speed becomes zero:

- There is deceleration / the resultant force is not zero, so the block is <u>not</u> in equilibrium; *[1 mark]*

(v) Calculate the magnitude of the change in momentum of the block:

List the known quantities:

- Initial velocity of block, u = 0.25 m s^{-1}
- Final velocity of block, v = −0.15 m s^{-1}

Calculate the change in momentum:

- $p = mv$ **or** $mu = 0.40 \times 0.25$ **or** $mv = 0.40 \times 0.15$ *[1 mark]*

- $\Delta\rho$ = momentum before − momentum after
- $\Delta\rho = mu - mv = mu - v$
- $\Delta\rho = 0.4 \times 0.25 - -0.15$
- $\Delta\rho = 0.16$ N s *[1 mark]*

[Total: 10 marks]

- *In part (v) "moves back along its original path" tells you that the block's **direction** is **opposite***
    - *So, the **velocity** is **negative***

Question 5 Medium [10 marks]

Part (a) [1 mark]

Define work done by a force.

Mark Scheme (1 marks):

The definition of work done:

- Force × displacement in the direction of the force; *[1 mark]*

[Total: 1 mark]

You will see mark schemes allowing 'distance travelled' or 'distance moved' instead of displacement in this defintion.

However, the definition above is the one given in your specification, and you should learn it this way.

Part (b) [3 marks]

A heavy suitcase is placed on a weighing machine consisting of a platform mounted on a spring as shown in Fig. 1.1.

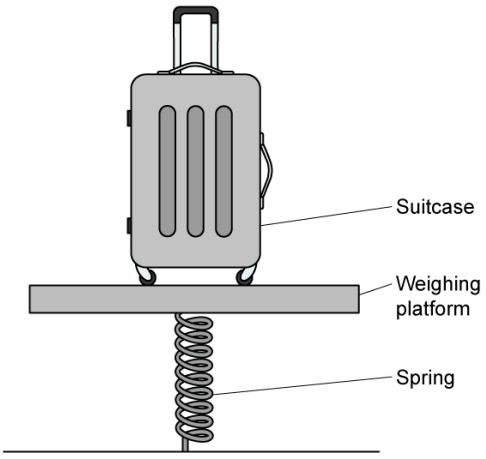


Fig. 1.1

Describe and explain how work is done in this situation.

Mark Scheme (3 marks):

How work is done when a heavy suitcase is placed on a spring:

- The weight (of the case) acts downwards and provides the force on the spring; *[1 mark]*
- The weight of the case compresses the spring by a distance; *[1 mark]*
- Work is done to compress the spring; this energy is stored as elastic potential energy in the spring; *[1 mark]*

[Total: 3 marks]

Part (c) [2 marks]

The diagram in Fig.1.2 shows a suitcase of mass 22.6 kg positioned on the weighing platform.

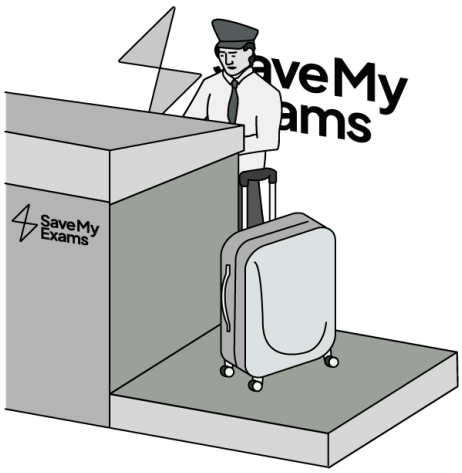


Fig. 1.2

The platform sinks down 3 cm when the suitcase is placed on it, and then returns to its original position when the suitcase is removed.

Calculate the spring constant of the spring.

Mark Scheme (2 marks):

Calculate the spring constant of the spring:

List the known quantities:

- Mass of the suitcase, m = 22.6 kg
- Compression of the spring, x = 3 cm = 0.03 m

Calculate applied force by finding the weight of the suitcase:

- $W = mg = 22.6 \times 9.81 = 222$ N *[1 mark]*

Use the equation for Hooke's Law to find the spring constant:

- $F = k\ x$
- $k = \frac{F}{x} = \frac{222}{0.03} = 7400$ Nm^{-1} *[1 mark]*

[Total: 2 marks]

Part (d) [4 marks]

For the weighing platform described in part (c) sketch a graph to show the known properties of the spring.

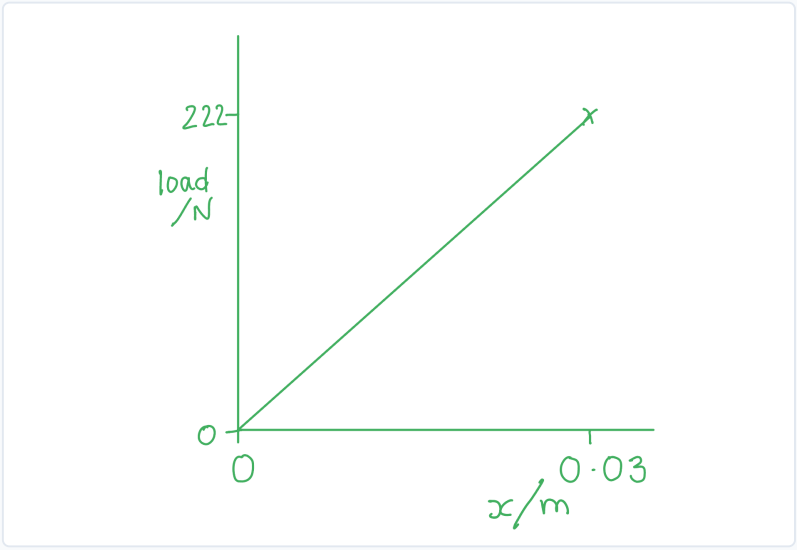
Mark Scheme (4 marks):

For the weighing platform described in part (c) sketch a graph to show the known properties of the spring:

Review the description of the weighing platform:

- When load = 222 N, extension = 0.03 m
- Platform returns to original position when applied load of 222 N is removed
- No data for heavier loads

Therefore the graph looks like:



- Both axes labelled with variable and unit; *[1 mark]*
- Line from (0, 0) to (0.03, 222); *[1 mark]*
- Line is straight (obeys Hooke's Law); *[1 mark]*
- Line does not extend beyond (0.03, 222); *[1 mark]*

[Total: 4 marks]

If you are asked to sketch the shape of a graph in a question that includes data, make sure to include that data on your sketch. Do not make up any data or assume any knowledge of what might happen if the investigation was continued.

Question 6 Medium [9 marks]

Part (a) [2 marks]

Nichrome is a common alloy used to make coins and heating elements in electrical appliances. It consists of 80% by volume of nickel and 20% by volume of chromium.

Determine the mass of nickel and the mass of chromium required to make a wire of nichrome of volume 0.50 × 10^{-3} m^3.

Density of nickel = 8.9 × 10^3 kg m^{-3}

Density of chromium = 7.1 × 10^3 kg m^{-3}

Mark Scheme (2 marks):

Determine the mass of nickel and the mass of chromium required to make a wire of nichrome of volume 0.50 × 10^{-3} m^3

Determine how much of the total volume will consist of each metal:

- Volume of nickel = 80% × total volume
- Volume of nickel = $0.8 \times (0.50 \times 10^{-3}) = 4.0 \times 10^{-4}$ m^3
- Volume of chromium = 20% × total volume
- Volume of chromium = $0.2 \times (0.50 \times 10^{-3}) = 1.0 \times 10^{-4}$ m^3

Use the equation for density to find the mass required:

- $m = \rho \times v$
- Mass of nickel, $m_n = \rho_n \times v_n = 8.9 \times 10^3 \times 4.0 \times 10^{-4} = 3.56$
- Mass of nickel, $m_n = 3.6$ kg 2 s.f. *[1 mark]*
- Mass of chromium, $m_c = \rho_c \times v_c = 7.1 \times 10^3 \times 1.0 \times 10^{-4} = 0.71$
- Mass of chromium, $m_c = 0.71$ kg 2 s.f. *[1 mark]*

[Total: 2 marks]

Part (b) [4 marks]

To calculate the breaking stress of the nichrome wire both the cross-sectional area of the wire and the applied load must be known.

For each value, briefly describe an experimental method to determine the value, mentioning any equipment required.

(i) The cross-sectional area of the wire.

[2]

(ii) The applied load.

[2]

Mark Scheme (4 marks):

An experimental method and equipment:

(i) The cross-sectional area of the wire:

- Use (Vernier) caliper / micrometer to find diameter; *[1 mark]*
- Measure in (at least) three places and take the average / mean; *[1 mark]*

(ii) The applied load:

- Use digital scales / balance to find mass; *[1 mark]*
- Multiply by *g* / 9.81 to find weight **OR** W = mg **OR** F = mg; *[1 mark]*

  OR

- Use force / Newton-meter; *[1 mark]*
- to find weight; *[1 marks]*

[Total: 4 marks]

Part (c) **[3 marks]**

The diameter of the nichrome wire is 5.7 mm. The wire breaks under an applied force of 72 kN.

Calculate the breaking stress of nichrome.

Mark Scheme (3 marks):

Calculate the breaking stress of nichrome

List the known quantities:

- Force, F = 72 kN = 72 000 N
- Diameter of the wire, d = 5.7 mm = 5.7×10^{-3} m

Determine the cross-sectional area of the wire:

- Area of a circle, $A = \pi r^2 = \pi \frac{d}{2}^2 = \pi \times \frac{5.7 \times 10^{-3}}{2}^2$ *[1 mark]*
- $A = 2.55 \times 10^{-5}$ m^2

Use the equation for stress:

- $\sigma = \frac{F}{A} = \frac{72\ 000}{2.55 \times 10^{-5}} = 2.82 \times 10^{9}$ Pa *[1 mark]*

Write answer to two sig figs and include the unit:

- $\sigma = 2.8 \times 10^9$ Pa *[1 mark]*

[Total: 3 marks]

Question 7 Hard [13 marks]

Part (a) [3 marks]

A student adds a series of masses to a vertical metal wire with a circular cross–section and measures the extension of the wire produced. This is shown in Fig. 1.1.

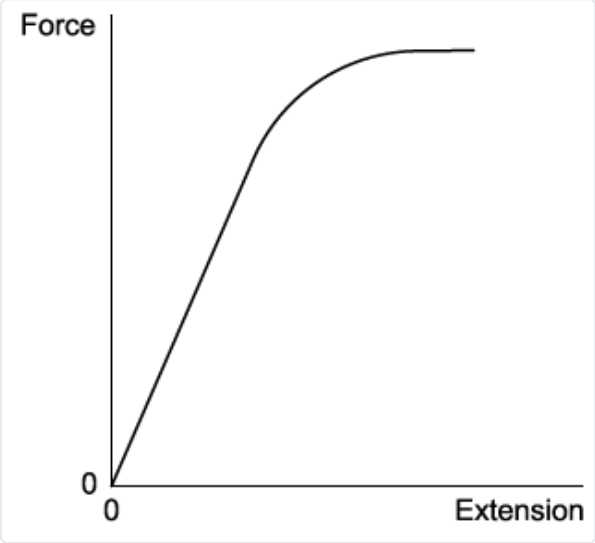


Fig. 1.1

Suggest how the student can determine the Young Modulus of the wire using the graph in Fig. 1.1.

Mark Scheme (3 marks):

Measurements the student needs to make:

- Young Modulus = $\frac{tensile\ \ stress}{tensile\ \ strain} = \frac{F \times L}{A \times \Delta L}$
- Therefore, the student needs to measure the original length **AND** the diameter of the wire; *[1 mark]*

Measurements from the graph:

- The gradient is equal to $\frac{F}{\Delta L}$ and then substitute that into the equation below; *[1 mark]*
- Calculate the Young Modulus using the equation $\frac{F \times L}{\pi \frac{d^2}{2} \times \Delta L}$ *[1 mark]*

[Total: 3 marks]

In your answer, you are allowed to say convert the results into a stress-strain graph and determine the gradient.

Part (b) [3 marks]

The metal wire is used to make a cable of diameter 4.9 mm. The Young Modulus of the metal in the cable is 154 GPa.

Calculate the force necessary to produce a strain of 0.40 % in the cable.

Mark Scheme (3 marks):

List the known quantities:

- The Young Modulus = 154 GPa = 154 × 10^9 Pa
- Diameter of the cable, d = 4.9 mm = 4.9 × 10^{-3} m
- Percentage strain = 0.40%

Calculate the stress on the wire:

- Young Modulus = $\frac{stress}{strain}$ therefore, Stress = Young Modulus × Strain
- 0.40% is equal to a strain of 0.004
- Stress = (154 × 10^9) × 0.004 = 6.16 × 10^8 Pa *[1 mark]*

Calculate the radius of the cable, r :

- $r = \frac{d}{2} = \frac{4.9 \times 10^{-3}}{2}$
- r = 2.45 × 10^{-3} m

Calculate the force, F :

- Stress = $\frac{F}{A}$ therefore, F = stress × A = stress × πr^2
- F = (6.16 × 10^8) × π × (2.45 × 10^{-3})^2 = 11 616.16 N *[1 mark]*
- F = 12 000 N **OR** 12 kN (2 s.f.) *[1 mark]*

[Total: 3 marks]

Remember to check you enter the correct value of the diameter or radius in your calculation for the cross-sectional area i.e. $A = \pi r^2$ *or* $A = \pi \frac{d^2}{2}$

Part (c) **[3 marks]**

The cable is used in a crane to lift a mass of m kg which is a third of the maximum possible mass it can lift.

Determine the maximum acceleration of the crane when it is lifting an object of mass m kg if the strain in the cable does not exceed 0.40%.

Mark Scheme (3 marks):

List the known quantities:

- Maximum force, F = 11 616.16 N (from part (b))
- Mass, $m = \frac{1}{3}M$ for maximum mass M

Method 1: Substitution

Determine the maximum acceleration, a_{max}:

Obtain an expression in terms of Newton's second law for mass m and maximum mass M:

- At maximum load $F_{max} = Mg$ for maximum mass M
- For mass m an expression for the maximum acceleration a_{max} is $F = ma_{max} = F_{max}$
- So, $ma_{max} = Mg$ *[1 mark]*

Obtain an expression for a_{max} in terms of acceleration due to free fall:

- Hence, $\frac{M}{3}a_{max} = Mg$
- $a_{max} = 3g$ *[1 mark]*

Calculate the value of a_{max}:

- $3g = 3 \times 9.81$
- $a_{max} = 29.43$ m s^{-2} *[1 mark]*

Method 2: Calculating masses

Calculate the maximum mass, M :

- $F_{max} = 11\,616.16$ N
- $F_{max} = Mg$
- $11\,616.16 = 9.81M$
- $M = 1184.11$ kg *[1 mark]*

Calculate the mass m:

- $m = \frac{1}{3}M$
- $m = \frac{1}{3} \times 1184.11$
- $m = 394.7$ kg *[1 mark]*

Calculate the maximum acceleration, a_{max}:

- From Newton's second law, $F = ma_{max}$
- $a_{max} = \frac{F}{m} = \frac{11616.16}{394.7} = 29.43$ m s^{-2} *[1 mark]*

[Total: 3 marks]

You have already considered the strain of 0.40% by using the force calculated from part b, so you don't need to apply this again.

Part (d) **[4 marks]**

Two different cables with spring constants k_1 and k_2 are connected in series.

Starting from Hooke's law, show that the combined spring constant, *K* of the two springs is given by the equation $K = \frac{k_1 k_2}{k_1 + k_2}$.

Mark Scheme (4 marks):

Consider the situation and obtain equations in terms of the force applied, *F*:

- Two springs connected in series provide a longer, flimsier spring
- When a force is applied to the combined spring the same force is applied evenly to each individual spring
- The springs have different spring constants so the extension they undergo is different
- Hence when x_1 is the extension for spring k_1 and x_2 is the extension of spring k_2:
    - $F = K(x_1 + x_2)$ (Equation 1)

      AND
    - $F = k_1 x_1$ (Equation 2)

      AND
    - $F = k_2 x_2$ (Equation 3) *[1 mark]*

Combine equations 2 and 3 to make x_1 the subject:

- $k_1 x_1 = k_2 x_2$
- $x_1 = \frac{k_2 x_2}{k_1}$ *[1 mark]*

Substitute for x_1 in equation 1:

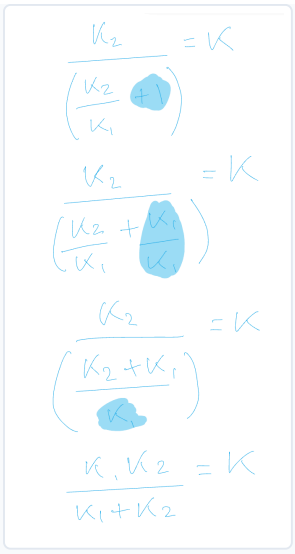
- $F = K\frac{k_2 x_2}{k_1} + x_2$
- $F = K\frac{k_2}{k_1} + 1 \; x_2$ *[1 mark]*

Combine this equation with equation 3 and rearrange to make *K* the subject:

- $k_2 x_2 = K\frac{k_2}{k_1} + 1 \; x_2$
- $k_2 = K\frac{k_2}{k_1} + 1$
- $\frac{k_2}{\frac{k_2}{k_1} + 1} = K$ *[1 mark]*

[Total: 4 marks]

Take your time and work through this type of question slowly thinking about the force applied when two wires are in series. You won't get marks for it because the question asks you ***to show*** *but you will need to check you have completed the question correctly by rearranging this final equation for K as follows:*


$\frac{K_2}{\left(\frac{K_2}{K_1}+1\right)} = K$
$\frac{K_2}{\left(\frac{K_2}{K_1}+\frac{K_1}{K_1}\right)} = K$
$\frac{K_2}{\left(\frac{K_2+K_1}{K_1}\right)} = K$
$\frac{K_1 K_2}{K_1+K_2} = K$

Question 8 Hard [18 marks]

Part (a) [6 marks]

As part of a quality check, a manufacturer of dog leads selects a sample of the wires used for the leads to a tensile test.

The sample of wires used is 3.0 m long and is of constant circular cross–section of diameter 0.92 mm. Hooke's law is obeyed up to the point when the wire has been extended by 20 mm at a tensile stress of 4.6×10^8 Pa.

The maximum load the wire can support before breaking is 350 N at an extension of 12 cm.

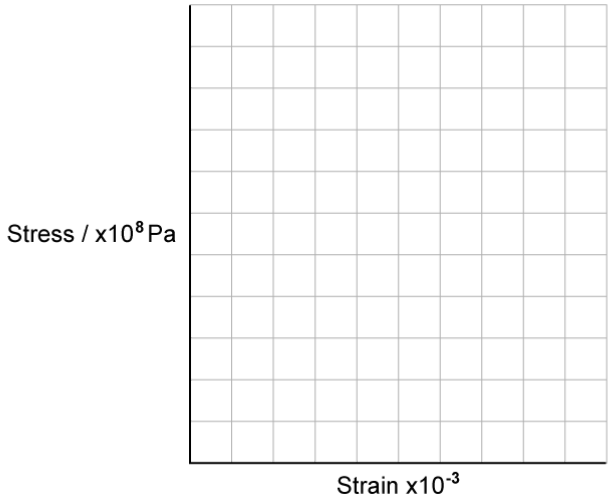


Fig. 1.1

Sketch a graph on Fig. 1.1 to show how you expect the tensile stress to vary with strain and mark the values of stress with corresponding strain at the limit of Hooke's law and the breaking point.

Mark Scheme (6 marks):

List the known quantities:

- Original length, L = 3.0 m
- Diameter, d = 0.92 mm = 0.92×10^{-3} m
- Tensile stress = 4.6×10^8 Pa
- Extension for Hooke's law limit = 20 mm = 0.02 m
- Breaking force, F = 350 N
- Breaking extension = 12 cm = 0.12 m

Calculate cross-sectional area, A:

- $A = \pi r^2 = \frac{\pi d^2}{4} = \frac{\pi \times 0.92 \times 10^{-3}{}^2}{4}$
- $A = 6.65 \times 10^{-7}$ m^2

Calculate the breaking stress:

- Breaking stress = $\frac{breaking\ \ force}{area}$ = $\frac{350}{6.65 \times 10^{-7}}$
- Breaking stress = 5.26 × 10^{8} Pa *[1 mark]*

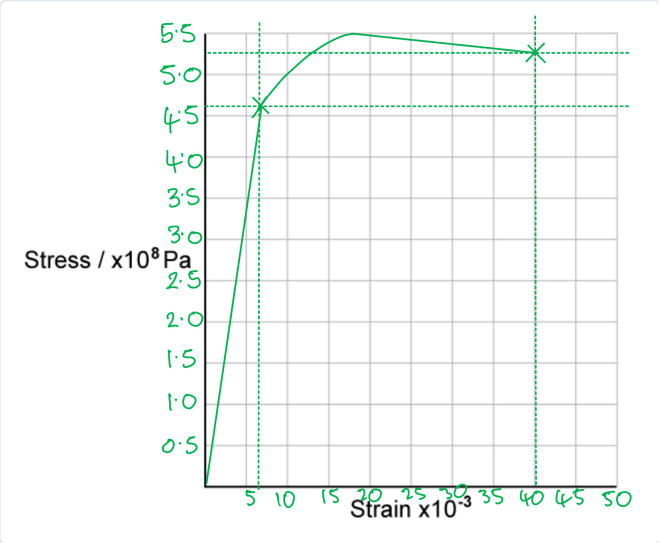
Calculate the breaking strain:

- Breaking strain = $\frac{Breaking\ \ extension}{L}$ = $\frac{0.12}{3}$ = 0.040 = 40 × 10^{-3} *[1 mark]*

Calculate the tensile strain:

- Tensile strain = $\frac{Extension\ \ for\ \ Hooke's\ \ law\ \ limit}{L}$ = $\frac{0.02}{3}$ = 0.0067 = 6.7 × 10^{-3} *[1 mark]*

A graph that would score 3 marks should look like:



- The graph line is a straight section up to the Hooke's law limit and is curved in the tensile section of the graph; *[1 mark]*
- Point at tensile stress and tensile strain (6.7 × 10^{-3}, 4.6 × 10^{8}); *[1 mark]*
- Point at breaking stress and breaking strain (40, 5.26 × 10^{8}); *[1 mark]*

[Total: 6 marks]

There are a few clues in the question as to the shape of this graph. 'Hooke's law is obeyed up to the point' means that the stress–strain curve will only be a straight diagonal line up to a point. Beyond this point, if it doesn't obey Hooke's Law this means the line starts to curve (i.e., The material starts to plastically deform). The 'maximum load before breaking' implies that at this stress–strain point, this graph will end. Therefore, the two points to calculate on the graph are the stress–strain at the end of the Hooke's law section, and the final stress–strain before the wire breaks.

Remember to label your axes appropriately, including all the units and any powers of 10 (e.g., × 10^{8}).

Part (b) **[4 marks]**

A different 3.0 m dog lead sample has a larger diameter. It is made of the same material and is subject to the same force.

Compare the advantages and disadvantages of this dog lead compared to the first.

Mark Scheme (4 marks):

Advantages:

- The second dog lead has a larger cross–sectional area, *A* **AND** since stress $=\frac{Force}{Area}$, it has a smaller tensile stress; *[1 mark]*
- It has the same Young Modulus, therefore, there is a smaller extension for the same force; *[1 mark]*
- For a given force, there is a smaller extension therefore the lead can experience a greater force before breaking; *[1 mark]*

Disadvantages:

*Any **one** from:*

- The dog lead will be heavier / harder to hold over long periods of time); *[1 mark]*
- The dog would be restricted by the lead; *[1 mark]*
- It is more difficult to make a knot in / wind up / change its shape; *[1 mark]*

[Total: 4 marks]

It may not be enough to get the mark by just writing 'stress and area are inversely proportional' although it is encouraged to state this. Remember to go on further and explain 'and therefore if the area increases, the stress decreases (for the same applied force), since stress = F/A'.

Part (c) **[4 marks]**

Calculate the force needed to extend a piece of 3.0 m leather dog leash with a radius of 2.3 mm to change its surface area by 4.1×10^{-3} m^2. Assume that the cross–sectional area is kept constant throughout the extension.

Young Modulus of leather = 0.05 GPa.

Mark Scheme (4 marks):

List the known quantities:

- Original length, L = 3.0 m
- Change in surface area = 4.1×10^{-3} m^2
- Radius, r = 2.3 mm = 2.3×10^{-3} m
- Young Modulus of leather = 0.05 GPa = 0.05×10^{9} Pa

Determine the equation for extension ΔL from the change in surface area, ΔSA:

- Surface area of a cylinder = $2\pi r L + 2\pi r^2$

- $\Delta SA = (2\pi r L_2 + 2\pi r^2) - (2\pi r L_1 + 2\pi r^2)$
- $\Delta SA = 2\pi r (L_2 - L_1) = 2\pi r \Delta L$
- $\Delta L = \frac{\Delta SA}{2\pi r}$ *[1 mark]*

Substitute into Young Modulus equation:

- Young Modulus = $\frac{FL}{A\Delta L} = \frac{FL}{A \times \frac{\Delta SA}{2\pi r}}$
- Cross-sectional area, $A = \pi r^2$
- Young Modulus = $\frac{FL}{\pi r^2 \times \frac{\Delta SA}{2\pi r}} = \frac{FL}{r \times \frac{\Delta SA}{2}} = \frac{2FL}{r \times \Delta SA}$ *[1 mark]*

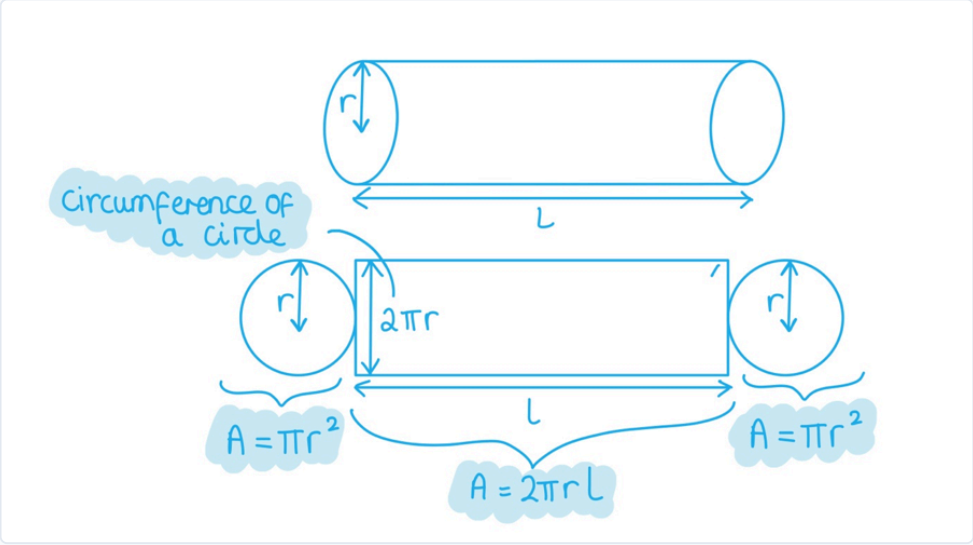
Calculate force, *F*:

- $F = \frac{Young \quad modulus \times r \times \Delta SA}{2L}$
- $F = \frac{0.05 \times 10^9 \times 2.3 \times 10^{-3} \times 4.1 \times 10^{-3}}{2 \times 3.0}$ *[1 mark]*
- $F = 78.58 = 79$ N *[1 mark]*

[Total: 4 marks]

If you forget the formula for areas and volumes for 3D shapes, don't panic! Just take time to figure them out from your GCSE Maths knowledge. For example: the curved surface area of the cylinder = 2πrh where r is the radius and h is the height of the cylinder.

The total surface area of a cylinder includes the areas of the two circular cross–sectionals on either side. To determine the surface area of any 3D shape, add up the area of all its faces. This is shown for the cylinder below:



Notice that the length of the shorter side of the rectangle that makes up the curved part of the cylinder is the circumference of a circle. The total surface area is therefore:

$$2\pi r L + \pi r^2 + \pi r^2 = 2\pi r L + 2\pi r^2$$

Part (d) **[4 marks]**

To prevent the walls of old buildings collapsing a metal rod is often used to tie opposite walls together, as shown in Fig. 1.2 below.

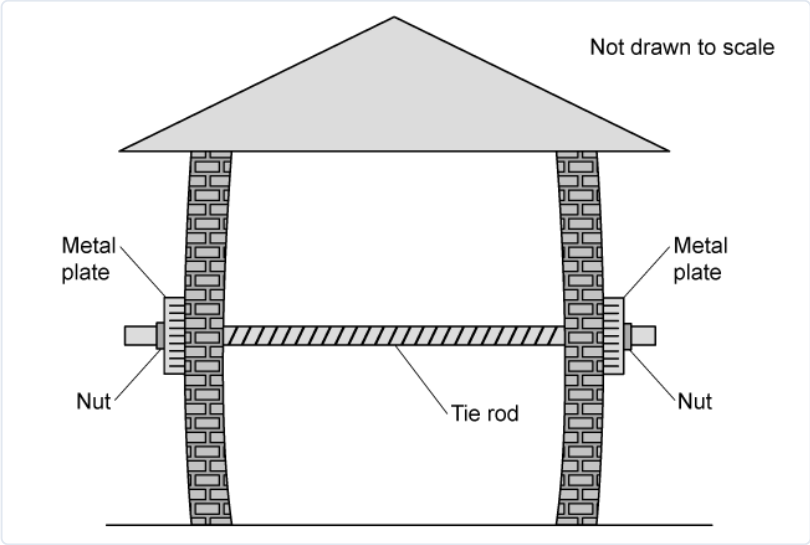


Fig. 1.2

In one case a steel tie rod of diameter 15 mm is used. When the nuts are tightened, the rod extends by 6 %. The Young Modulus of steel is 2.5×10^{11} Pa.

Calculate the force exerted on the walls by the rod.

Mark Scheme (4 marks):

List the known quantities:

- Diameter, d = 15 mm = 15×10^{-3} m
- Young Modulus of steel = 2.5×10^{11}Pa
- Extension of rod = 6 %

Calculate the strain:

- Strain = $\frac{\Delta L}{L}$ = 6 % = 0.06 *[1 mark]*

Calculate the cross-sectional area, A:

- $A = \pi r^2 = \frac{\pi d^2}{4}$
- $A = \frac{\pi \times 15 \times 10^{-3\ 2}}{4}$ = 1.77×10^{-4} m^2 *[1 mark]*

Calculate the stress:

- Young Modulus = $\frac{stress}{strain}$ therefore, Stress = Strain × Young Modulus
- Stress = 0.06 × (2.5×10^{11}) = 1.50×10^{10} Pa *[1 mark]*

Calculate force, F:

- Stress = $\frac{F}{A}$ therefore, F = Stress × A

- $F = (1.50 \times 10^{10}) \times (1.77 \times 10^{-4})$
- $F = 2.66 \times 10^{6}$ N = 2.66 MN *[1 mark]*

[Total: 4 marks]

*If the rod extends by 6 % that means the **ratio** of it extension to the original length is equal to 6 % or 0.06. You can check this by putting in some numbers:*

For example:

- *If L = 5 m, therefore L + ΔL = 5 × 1.06 = 5.3 (increased by 6 %)*
- *Strain* $= \frac{\Delta L}{L} = \frac{5.3 - 5}{5} = 0.06$

*This force is the force exerted **on** the rod **by** the walls, but according to Newton's third law, this is equal in size to the force exerted **by** the rod **on** the walls.*